

Appendix C - Eastern Recherche Marine Park Data Analysis Report

South-west Marine Parks Network

Benthic ecosystem extent

Table C 1.1: Habitat model selection results. Top generalised additive models (GAMs) for predicting the probability of occurrence of key habitats from full subset analyses. Differences between the lowest reported corrected Akaike Information Criterion (ΔAICc), AICc weight (ωAICc), variance explained (R^2) and effective degrees of freedom (EDF) are reported for model comparison. Model selection was based on the most parsimonious model (fewest variables - shown in bold) within two units of the lowest AICc.

Response	Model	ΔAICc	ωAICc	R^2	EDF
Macroalgae	detrended+roughness+Z	0	1	0.68583	6.99
Sand	detrended+roughness+Z	0	1	0.41013	6.60
Sessile invertebrates	detrended+roughness+Z	0	1	0.06906	6.92
Reef	detrended+roughness+Z	0	1	0.42445	5.00

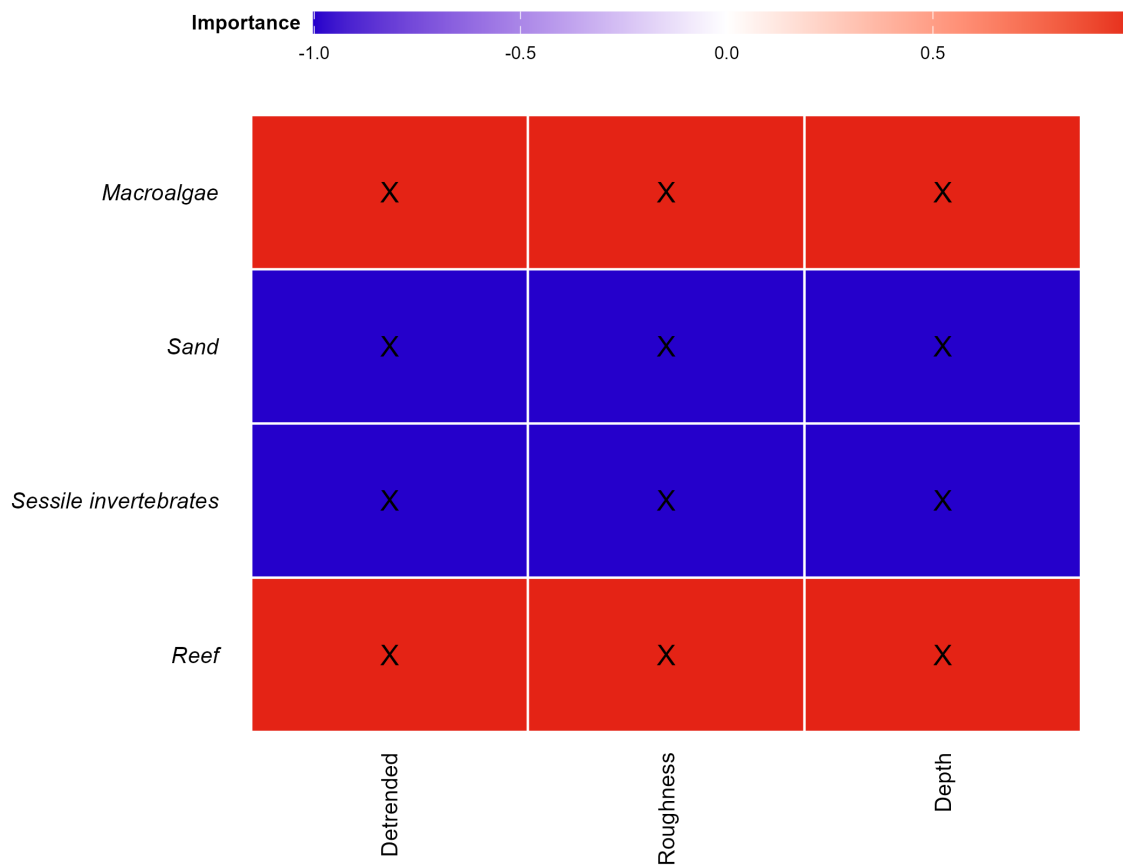


Figure C 1.1: Variable importance scores for predicting the probability of occurrence of main habitat categories. Response variables included in the most parsimonious top model are indicated (X), with the colour gradient representing positive (red), zero (white) and negative (blue) relationships.

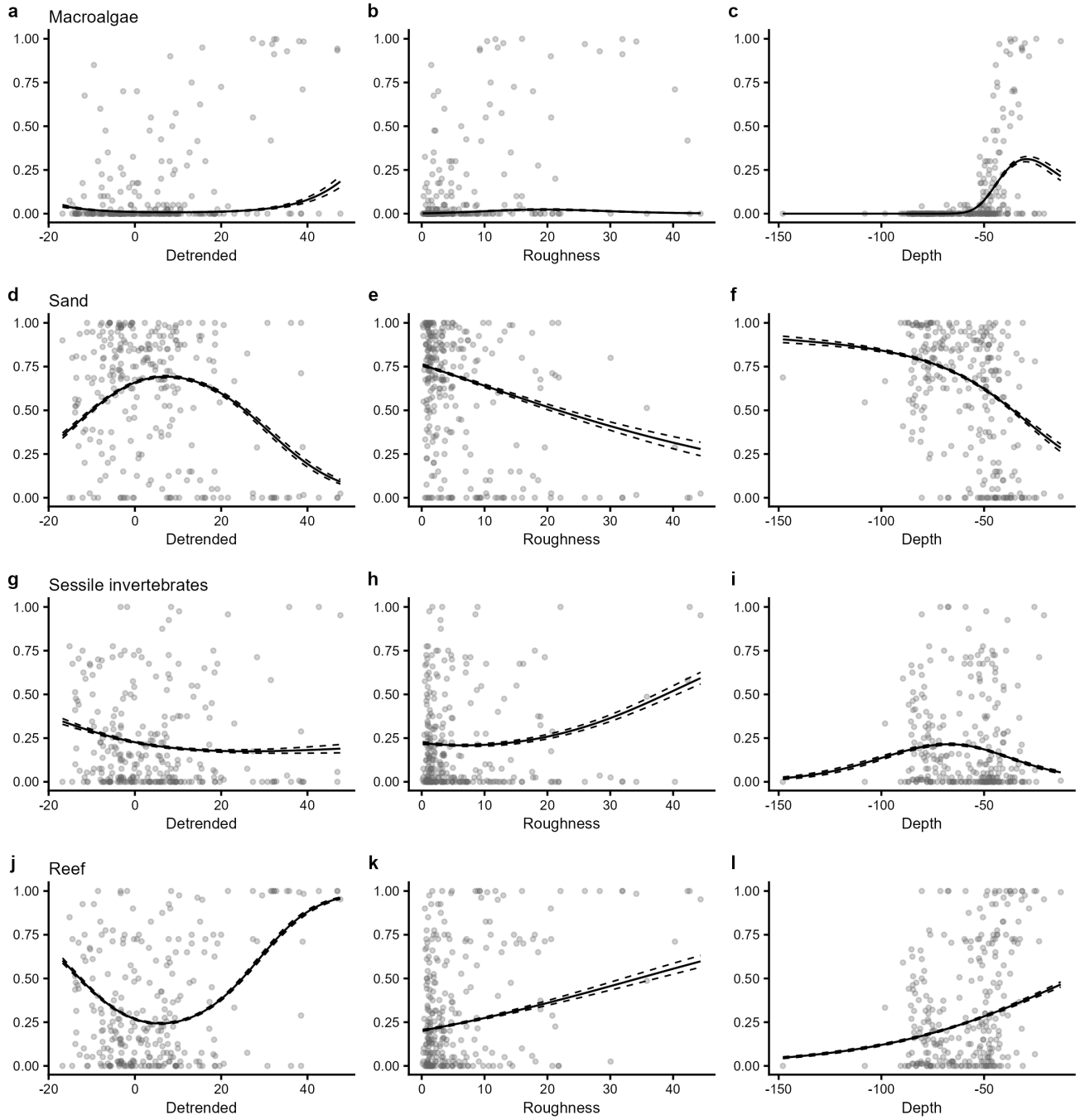


Figure C 1.2: Plots of the most parsimonious model found to predict the probability of occurrence of key habitat types. Individual panes display the predicted relationships between model covariates and the probability of occurrence of each habitat class. Solid lines represent predicted Generalized Additive Model (GAM) fits with other variables held constant at their mean value. Dashed lines represent upper and lower standard error bounds around the prediction, and points represent the observed data.

Demersal fish assemblage

Table C 2.1: Demersal fish model selection results. Top generalised additive models (GAMs) for predicting the species and abundance distribution of metrics of interest from full subset analyses. Differences between the lowest reported corrected Akaike Information Criterion (ΔAICc), AICc weight (ωAICc), variance explained (R^2) and effective degrees of freedom (EDF) are reported for model comparison. Model selection was based on the best model (fewest variables - shown in bold) within two units of the lowest AICc. *Large Reef Fish Index representing the biomass of all fish greater than 20 cm (B20) has been adapted here for stereo-BRUV data by removing sharks and rays.

Response	Model	ΔAICc	ωAICc	AICc	R^2	EDF
Species richness	roughness+reef+year	0	0.306	682.78	0.69245	4.86
	reef+year	0.113	0.29	682.90	0.68819	3.84
	detrended+reef+year	1.523	0.143	684.30	0.6931	5.46
	reef	1.691	0.132	684.47	0.68133	2.85
Total abundance	roughness+reef	0	0.572	1,460.92	0.46629	3.74
	roughness+reef+year	1.799	0.233	1,462.72	0.46771	4.75
Large Reef Fish Index*	detrended+reef	0	0.573	1,999.80	0.21359	3.68
	detrended+reef+year	0.602	0.424	2,000.41	0.21673	4.24
Reef Fish Thermal Index	Z	0	0.441	298.84	0.21534	2.97
	detrended+reef+year	0.531	0.338	299.37	0.23039	4.37

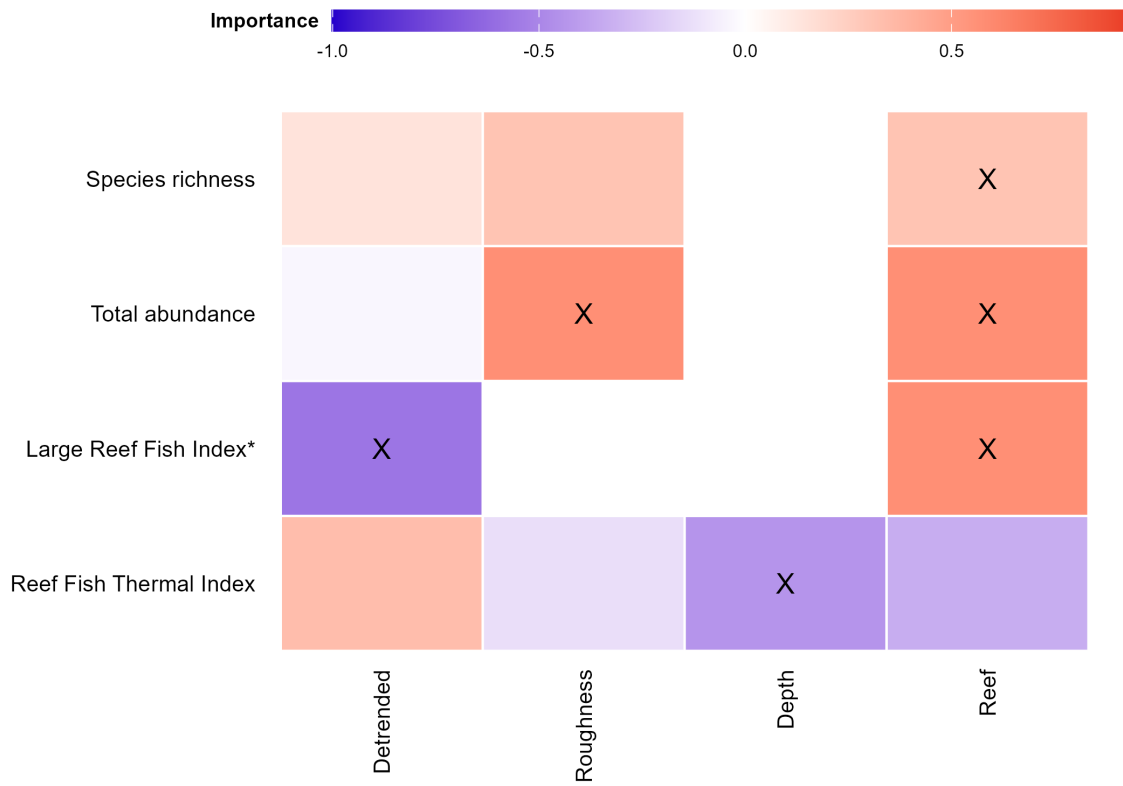


Figure C 2.1: Variable importance scores for predicting the probability of demersal fish metrics. Response variables included in the most parsimonious top model are indicated (X), with the colour gradient representing positive (red), zero (white) and negative (blue) relationships. *Large Reef Fish Index representing the biomass of all fish greater than 20 cm (B20) has been adapted here for stereo-BRUV data by removing sharks and rays.

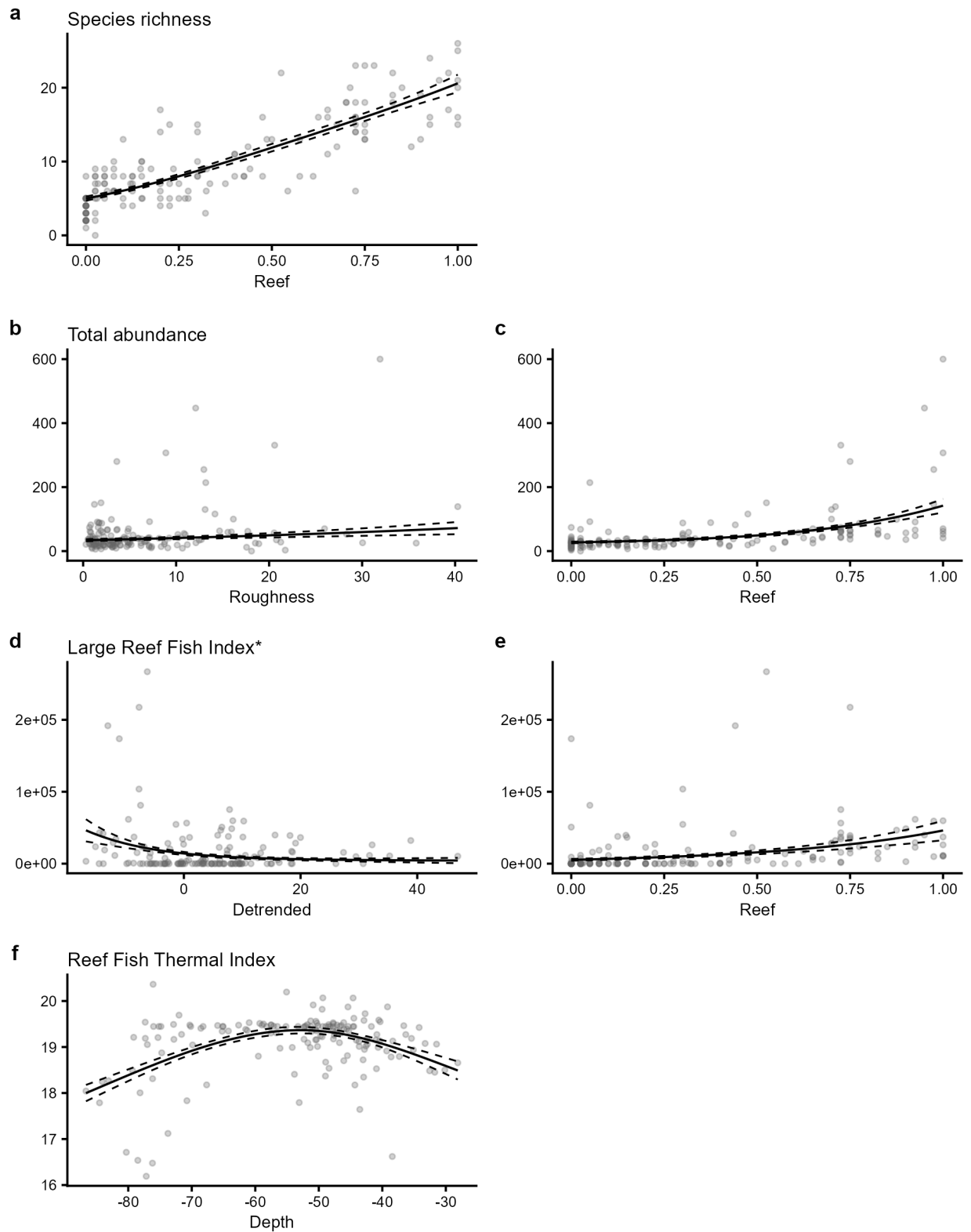


Figure C 2.2: Plots of the best model found to predict the probability of occurrence of key fish groups. Individual panes display the predicted relationships between model covariates and each demersal fish metric. Solid lines represent predicted Generalized Additive Model (GAM) fits with other variables held constant at their mean value. Dashed lines represent upper and lower standard error bounds around the prediction, and points represent the observed data. *Large Reef Fish Index representing the biomass of all fish greater than 20 cm (B20) has been adapted here for stereo-BRUV data by removing sharks and rays.